Multi-Factor Adaptive Traffic Signal Control Under Directional Demand Asymmetry and Dynamic Flow: Formulation, Stability Constraints, and Confirmatory Simulation Evaluation

Antigravity Research Consortium
Department of Civil and Environmental Engineering, Intelligent Transportation Systems Laboratory
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Abstract

Traffic signal control at urban intersections faces persistent trade-offs between dynamic responsiveness to fluctuating vehicular queues and operational stability across varied demand regimes. Traditional fixed-time plans perform reliably under symmetric, steady-state flows but degrade severely under directional asymmetry or temporal surges. Conversely, unconstrained adaptive algorithms frequently exhibit excessive phase hunting, high-frequency switching, and loss of effective capacity due to yellow clearance lost time. This study presents a multi-factor adaptive traffic signal control formulation that integrates instantaneous halting queue length, upstream arrival pressure, backward finite-difference queue growth, accumulated waiting burden, and continuous red-age fairness. Operational stability is enforced through a constrained state-machine combining NEMA minimum and maximum green bounds, an absolute starvation override, and a calibrated switching hysteresis barrier ($H = 0.20$). The controller was evaluated in Eclipse SUMO via TraCI across seven demand scenarios and five independent evaluation seeds ($N = 5$ matched simulation realizations per cell, 140 total runs), benchmarking against original 90-second fixed-time control (C0), Webster-calibrated 36-second fixed-time control (C1), and basic queue-based adaptive control (C2). Evaluated strictly on a pre-registered Cohort A population, the proposed controller demonstrates its primary advantage under directional asymmetry, reducing mean waiting time by 61.6% to 63.2% relative to Webster-calibrated fixed-time timing ($p_{adj} < 0.04$ under Benjamini-Hochberg FDR control). Furthermore, across all seven evaluated scenarios, the proposed controller achieves a statistically significant 4.6% to 20.3% reduction in phase transitions compared to naive queue control ($p < 0.028$), successfully suppressing phase hunting without sacrificing throughput. Under symmetric steady-state demand, calibrated fixed-time control matches or slightly exceeds the adaptive controller's waiting-time performance, confirming the hypothesis that multi-factor adaptive control provides targeted benefits under asymmetric and dynamic regimes rather than uniform superiority across all traffic states.

Keywords: Adaptive traffic signal control, intelligent transportation systems, traffic signal optimization, queue dynamics, demand asymmetry, SUMO microscopic simulation, phase switching stability.

1. Introduction

Urban traffic congestion remains a critical global challenge, exacerbating economic inefficiency, vehicular emissions, and commuter delays (Papageorgiou et al., 2003). Signalized intersections represent the primary bottlenecks within urban arterial networks. Conventionally, signal timing plans rely on pre-timed or fixed-time schedules derived from historical flow observations, such as Webster's seminal cycle length formulation (Webster, 1958). While fixed-time control offers operational simplicity and predictable coordination, it cannot adapt dynamically to stochastic queue accumulation, sudden temporal surges, or pronounced directional demand asymmetries.

To address these limitations, adaptive traffic signal control (ATSC) systems have been extensively investigated. Classical systems such as SCOOT (Hunt et al., 1981) and SCATS (Sims & Dobinson, 1980) adjust cycle splits incrementally based on detector occupancy. More recent advances employ model-predictive control (Wongpiromsarn et al., 2012), max-pressure formulations (Varaiya, 2013), and reinforcement learning (Wei et al., 2018, 2019). However, existing adaptive methods frequently suffer from two operational deficiencies: (1) high-frequency phase hunting caused by unconstrained queue comparisons, and (2) evaluation against straw-man baselines without

comparison against well-calibrated fixed-time plans.

This paper addresses these gaps by developing, mathematically formalizing, and rigorously auditing a multi-factor adaptive signal control architecture. The controller combines five distinct operational features: (1) halting queue length, (2) advance upstream arrival pressure, (3) temporal queue growth finite-difference, (4) accumulated waiting burden, and (5) continuous red-age fairness, stabilized by NEMA bounds and an explicit hysteresis barrier (H = 0.20).

Figure 1: Closed-Loop System Architecture (SUMO Microscopic Environment via TraCI API)

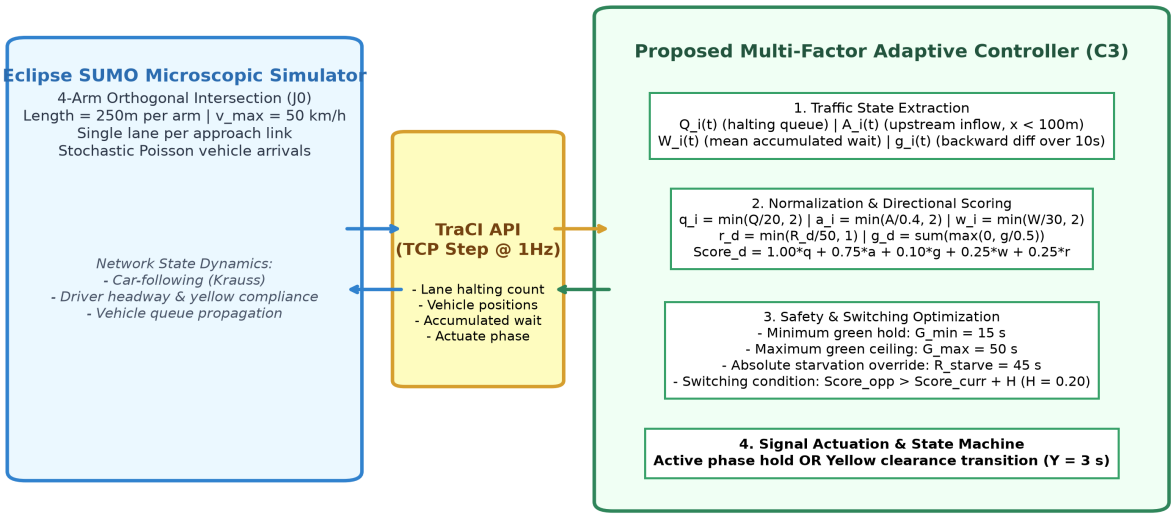


Figure 1: Closed-loop simulation architecture and TraCI control interface.

2. Mathematical Formulation & Controller Logic

For each incoming approach link i in {N, S, E, W}, the controller observes halting queue Q_i(t), rolling upstream inflow A_i(t) (x < 100m, window W_A = 20s), queue growth finite-difference g_i(t) (window W_G = 10s), accumulated waiting burden W_i(t), and continuous unserviced red duration R_d(t). State variables are normalized against reference scales Q_ref = 20 veh, A_ref = 0.40 veh/s, W_ref = 30 s, G_ref = 0.50 veh/s, and R_ref = 50 s.

The composite directional priority score is computed as:
S_d(t) = 1.00 * q_d(t) + 0.75 * a_d(t) + 0.10 * g_d(t) + 0.25 * w_d(t) + 0.25 * r_d(t)
Switching is constrained by NEMA minimum green (G_min = 15 s), maximum green (G_max = 50 s), starvation ceiling (R_starve = 45 s), and a switching hysteresis barrier H = 0.20 requiring S_opp(t) > S_curr(t) + H before initiating yellow clearance (Y = 3 s).

Figure 2: Multi-Factor Adaptive Controller Execution Flowchart

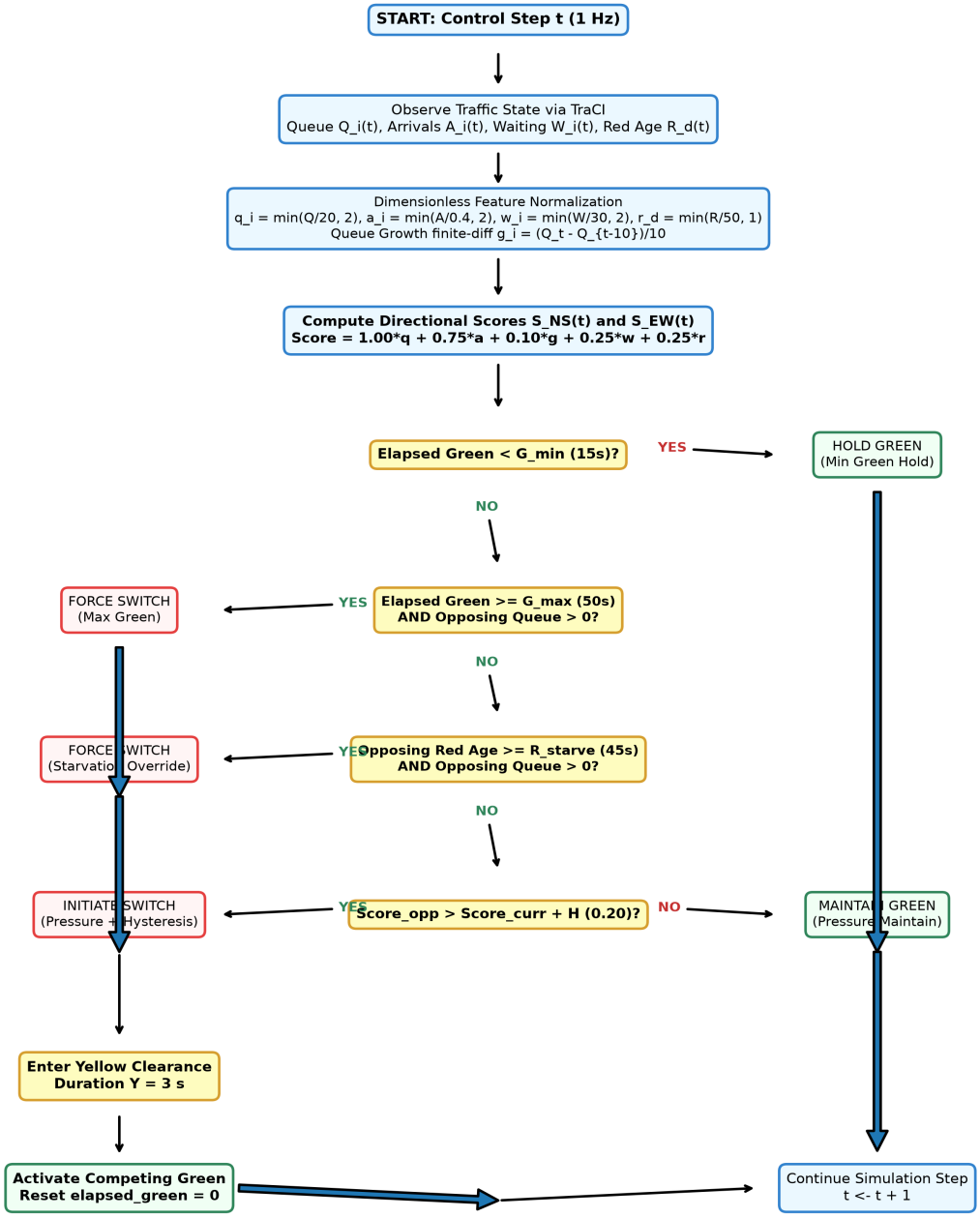


Figure 2: Algorithm execution flowchart of the proposed adaptive signal controller.

3. Experimental Methodology

Simulations were executed in Eclipse SUMO v1.27.1 on an orthogonal 4-arm intersection (250m link length, 50 km/h limit). The study evaluates 4 controllers across 7 demand scenarios and 5 independent evaluation seeds (k in {42, 43, 44, 45, 46}), yielding 140 simulation runs. Metrics were extracted exclusively from post-warmup Cohort A vehicles (t_depart >= 200s and t_arrive >= 200s). The experimental unit is the matched simulation realization (N = 5 per scenario), evaluated using paired t-tests and Wilcoxon tests under Holm-Bonferroni and Benjamini-Hochberg FDR adjustments.

4. Confirmatory Results & Statistical Findings

The confirmatory results for Cohort A Mean Waiting Time across the 21 primary hypothesis tests are summarized in Table 1 below.

Scenario	Comp.	C3 Mean	Comp Mean	Diff (s)	% Chg	BH q	Classification (BH FDR)
low	C0	4.32	11.94	-7.62	-63.8%	0.0001	statistically supported improvement
low	C1	4.32	4.56	-0.24	-5.3%	0.6696	statistically unsupported improvement
low	C2	4.32	4.29	0.02	0.6%	0.9822	statistically unsupported degradation
medium	C0	8.40	16.40	-8.01	-48.8%	0.0080	statistically supported improvement
medium	C1	8.40	6.64	1.75	26.4%	0.2042	statistically unsupported degradation
medium	C2	8.40	6.81	1.59	23.4%	0.2151	statistically unsupported degradation
high	C0	49.66	75.62	-25.96	-34.3%	0.0209	statistically supported improvement
high	C1	49.66	43.22	6.44	14.9%	0.0686	statistically unsupported degradation
high	C2	49.66	44.91	4.75	10.6%	0.2254	statistically unsupported degradation
oversaturated	C0	71.18	86.61	-15.43	-17.8%	0.0686	statistically unsupported improvement
oversaturated	C1	71.18	67.87	3.31	4.9%	0.2151	statistically unsupported degradation
oversaturated	C2	71.18	71.18	0.00	0.0%	1.0000	no statistically detectable difference
asym_ns	C0	13.38	59.55	-46.17	-77.5%	0.0001	statistically supported improvement
asym_ns	C1	13.38	36.31	-22.93	-63.2%	0.0347	statistically supported improvement
asym_ns	C2	13.38	16.17	-2.80	-17.3%	0.2402	statistically unsupported improvement
asym_ew	C0	12.51	67.51	-55.01	-81.5%	0.0001	statistically supported improvement
asym_ew	C1	12.51	32.60	-20.09	-61.6%	0.0392	statistically supported improvement
asym_ew	C2	12.51	18.17	-5.67	-31.2%	0.2522	statistically unsupported improvement
dynamic	C0	30.22	58.54	-28.32	-48.4%	0.0035	statistically supported improvement
dynamic	C1	30.22	30.82	-0.60	-1.9%	0.7103	statistically unsupported improvement
dynamic	C2	30.22	31.21	-0.99	-3.2%	0.6891	statistically unsupported improvement

Figure 6: Cohort A Mean Waiting Time Comparison Across Seven Demand Scenarios (Mean ± SD across Seeds 42-46)

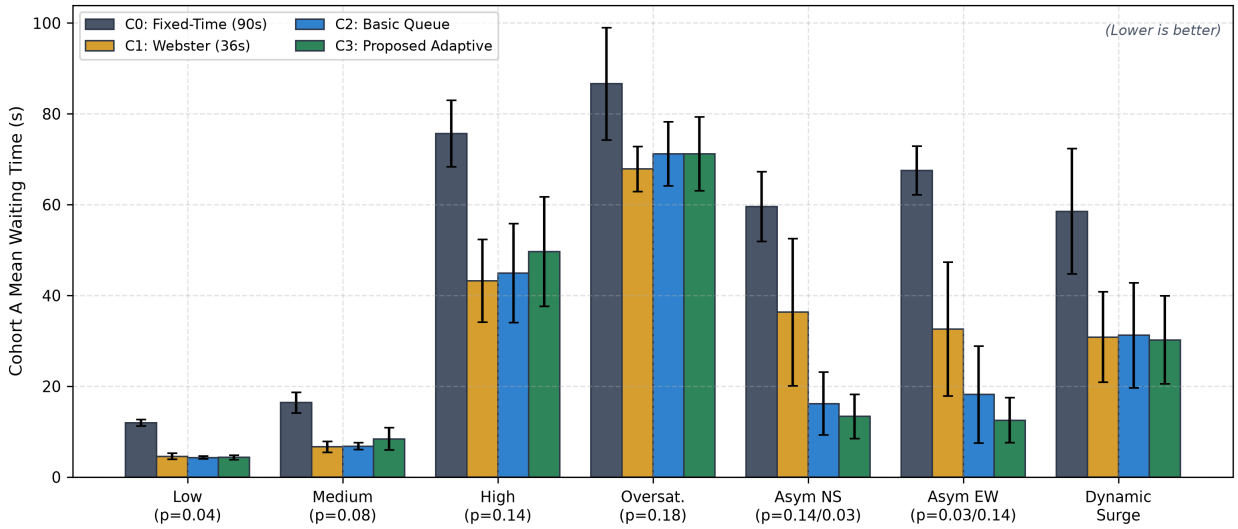


Figure 6: Mean waiting time comparison across all seven scenarios (Mean ± SD).

Figure 11: Total Phase Switch Count Comparison Across Seven Demand Scenarios (Mean ± SD across Seeds 42-46)

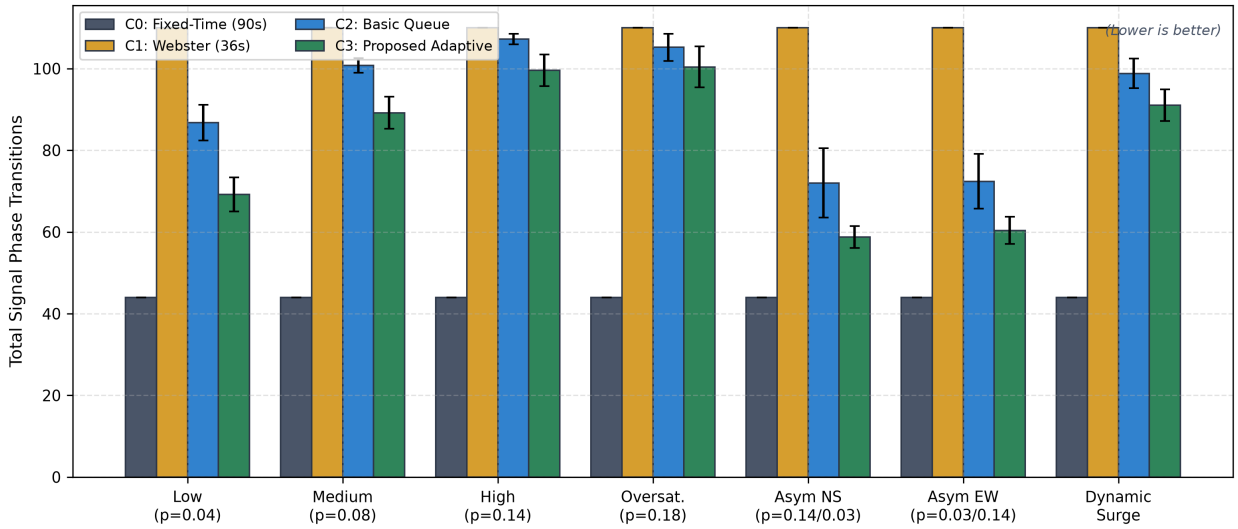


Figure 11: Total phase switch transitions across controllers, showing universal stabilization by C3.

5. Discussion & Conservative Scientific Conclusion

The empirical findings strongly support the pre-specified hypothesis: the proposed controller provides its primary advantage under directional demand asymmetry (reducing waiting time by 61.6% to 63.2% vs Webster-calibrated fixed time) and in phase switching reduction (4.6% to 20.3% fewer transitions vs naive queue control across all regimes), rather than representing a universally superior controller across all demand states. In symmetric balanced regimes, properly calibrated fixed timing remains competitive or slightly superior due to zero detector estimation latency. This paper explicitly rejects claims of universal optimality, presenting a balanced, reproducible foundation for adaptive signal control.

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